Fiscal Year:	FY 2021	**Task Last Updated:**	FY 06/09/2025
PI Name:	Poirier, David Sc.D.		
Project Title:	Effect of Varying Convection on Dendrite Morphology and Macrosegregation		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	poirierd@email.arizona.edu	**Fax:**	FY 520-621-8059
PI Organization Type:	UNIVERSITY	**Phone:**	520-621-6072
Organization Name:	The University of Arizona		
PI Address 1:	1235 East James E Rogers Way		
PI Address 2:	Department of Materials Science and Engineering		
PI Web Page:	http://www.mse.arizona.edu/		
City:	Tucson	**State:**	AZ
Zip Code:	85721-0012	**Congressional District:**	7
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	12/01/2014	**End Date:**	08/13/2021
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Grugel, Richard	**Contact Phone:**	256 544-9165
Contact Email:	Richard.N.Grugel@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/13/2021 per NSSC info (Ed., 6/25/21) NOTE: End date change to 4/30/2021 per NSSC information (Ed., 9/10/2020)		
Key Personnel Changes/Previous PI:	October 2017 report: Prof. D.R. Poirier, Dept. of Materials Science and Engineering, The University of Arizona, remains as the P.I. Dr. V. Manga, Assistant Research Professor, in the Dept. of Materials Science and Engineering at The University of Arizona has been the co-investigator since early 2016. During 2019, Dr. Manga was assisted by a post-doctoral researcher on a temporary basis. Her main responsibility was to introduce Dr. Manga to software (ANSYS-Fluent) for simulating macrosgregation.		
COI Name (Institution):	Manga, Venkat Ph.D. (The University of Arizona) Tewari, Surendra Ph.D. (Cleveland State University)		
Grant/Contract No.:	NNX15AC04G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE (Ed., 12/16/2014): This project is a continuation of grant NNX10AV40G, "Effect of Varying Convection on Dendrite Morphology and Macrosegregation," period of performance of 12/1/2010-11/30/2014, and with the same Principal Investigator, Dr. David Poirier. The research of this grant pertains to convection that is driven by directional solidification (DS) through changes in cross-sections (because of solidification shrinkage) and thermosolutal convection. In microgravity, of course, the thermosolutal convection will be mitigated so that effects of the shrinkage-driven flow in cross-sectional changes can be isolated from the usual thermosolutal convection that always occurs when solidification occurs on Earth. The outcome of the research is expected to be relevant to commercial DS-castings, which are used in the high-temperature sections of gas-turbine engines (i.e., jet engines and electric power generators). Such sophisticated castings have geometries, which invariably have changes in cross-sections. Yet convection at a cross-sectional change and its relationship to the formation of defects in DS-castings and/or in dendritic mono-crystals have not been studied well. Hence, the research not only has scientific merit, it is also relevant to the production of commercial DS castings. The research is on directional solidification experiments that are compatible with the solidification cartridge assemblies (SCAs) that can be employed in a furnace on the International Space Station (ISS). The research comprises: 1) selected alloys have varying density-responses to gravity forces; 2) crucible designs with variable cross-sections but compatible with the current SCAs; 3) sophisticated characterizations of the solidified microstructures, including both micro- and macro-segregation; and 4) supporting computer simulations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The outcome of the research is expected to be relevant to commercial directionally solidified-castings, which are used in the high-temperature sections of gas-turbine engines (i.e., jet engines and turbine electric power-generators). Such sophisticated castings have geometries, which invariably have changes in cross-sections. Yet convection at a cross-sectional change and its relationship to the formation of defects in DS-castings and/or in dendritic mono-crystals have not been studied well. Hence, the research is very much needed for its scientific merit and also because it is relevant to the production of commercial DS castings.
Task Progress:	June 2025 Update per the Principal Investigator (PI): The results of the simulations of the all-liquid zones and the mushy-zones reveal the thermal fields, the convection in the all-liquid zoned, the shapes of the mushy zones, and the macrosegregation. Ultimately, macrosegregation in the completely solidified casting is of interest to casting practitioners. • The macrosegregation revealed for mushy zones are taken to be indicative of macrosegregation in the solidified castings. • The macrosegregation for the cases considered in this investigation show the effect of using a graphite-mold versus an alumina-mold and the effect of changing the solidification rate. A manuscript was published in the Journal of Crystal Growth [Ed. Note: See Cumulative Bibliography.]
Bibliography Type:	Description: (Last Updated: 06/06/2025)
Articles in Peer-reviewed Journals	Ghods M, Upadhyay SR, Rajamure RS, Tewari SN, Grugel RN, Poirier DR, Lauer M. "Primary dendrite array morphology in Al-7 wt% Si alloy samples directionally solidified aboard the International Space Station." Journal of Crystal Growth. 2021 May 15;562:126077. https://doi.org/10.1016/j.jcrysgro.2021.126077 , May-2021